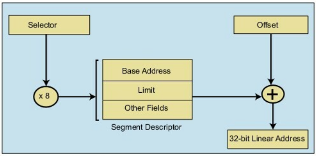

Figure 3: Segment translation

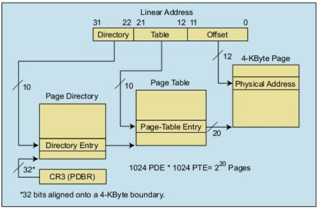

Figure 4: Page translation (from Intel's official documentation) (*Courtesy*: Intel website)

segment; thus, this is the segment limit.

- *Other control bits* – Some of the other control bits are:
  - *DPL*- Descriptor Privilege Level; the value 0 means the segment is in the kernel mode; however, the value 3 means it is in user mode.
  - *P bit* - Segment Present Bit; the value 0 means the segment is not present in the main memory and is swapped back to the disk.
  - *S bit* - Other than segment descriptors, the GDT can also hold things like the Task State Segment (TSS), the LDT or Call Gate descriptors. If this bit is cleared, the segment is one of the system segments like TSS, LDT or Call Gate; otherwise, it is a normal code or data segment.

Another type of descriptor table that x86-system has is the *Interrupt Descriptor Table, or IDT.* This data-structure is used to implement the correct response to hardware/software interrupts and processor exceptions, and thus is nothing but an *interrupt vector table.* A special register called *idtr* register contains the address of the IDT.

A few words about TSS before we discuss paging. As the name suggests, *Task State Segment Descriptor or TSS*[3] is used to save the state of a task. An operating system does a task switch with the help of TSS, and thus multitasks.

A simple page translation technique

When the paging unit is active, mapping of a linear address to a physical address involves a paging technique. When the PG bit of the control register CR0 is set, the paging unit is active, and thus the segment requires a second level of address translation. The PG bit is usually set by the operating system to implement page oriented protection.

Paging is used to separate the tasks and create an independent virtual address space for every task in the memory. As a key task, the paging unit checks the requested access type against the access rights of the linear address, and generates a page fault exception if the memory access is invalid. Physical continuous addresses are grouped into fixed length chunks called a *page or a physical page* (usually of 4K byte).

In an x86-system, a 32-bit linear address is divided into the following three fields:

- Directory
- Table
- Offset

Further, with the help of two translation tables, a linear address gets translated to a physical address:

- Page directory
- Page table

Each active process has a page directory associated to it. Control register CR3 (also known as the Page Directory Base Register, or PDBR) contains the base physical address of the page directory of the task.

The *Directory* field of a linear address is used to determine the correct page table entry from the *page* directory. Next, the Page field (of the linear address) helps in pointing the correct page frame entry from the page table. The *Offset* field then determines the relative position within the page frame.

Figure 4 (from Intel's official documentation) depicts the page translation diagrammatically.

I am sure you might have loads of questions, so Google around or even feel free to write to me. I will try to clear your doubts. END

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